

# LAG

Keep missing values and order meaningful

01 / THE SAS EXAMPLE

```
data test;
  input id value;
  prev_value = lag(value);
  prev2_value = lag2(value);
  change = value - prev_value;
  datalines;
1 10
2 15
3 12
4 18
5 20
;
run;
proc print data=test noobs;
run;
```

REPORTED TEST SETUP / 21 SEP 2026

Claude Sonnet 4.5 / AWS Bedrock GovCloud  
Claude Code 2.1.278 / effortLevel=high / prompt v0.1.0

ATTACH SOURCE OR PASTE IT AFTER THE PROMPT. START FRESH EACH TIME.

WHAT YOU WILL PRACTICE

State what the model should preserve when there is no previous value.

Each LAG call executes on every row in this program. The supplied order defines the previous values. Empty starting positions mean no earlier observation; they remain missing.

EXPECTED RESULT

| id | value | prev_value | prev2_value | change |
|----|-------|------------|-------------|--------|
| 1  | 10    | .          | .           | .      |
| 2  | 15    | 10         | .           | 5      |
| 3  | 12    | 15         | 10          | -3     |
| 4  | 18    | 12         | 15          | 6      |
| 5  | 20    | 18         | 12          | 2      |

A period denotes SAS numeric missing; pandas may display NaN or <NA>. Compare missing positions as well as numbers. Conditional LAG calls require a separate analysis.

Reported output matches: Prompt A 3/3 | Prompt B 3/3

Reference checked in Python; SAS execution pending.

# Prompt A: leave choices open

High degrees of freedom / Notice the decisions the model makes for you.

COPY THIS PROMPT + THE SAS EXAMPLE

Convert this SAS code to Python.  
Show code in this response. Do not create or run files, or read existing Python files.

REPORTED TRIAL RESULTS / 21 SEP 2026

## All A runs chose row shifts

| Run | Method              | Output match |
|-----|---------------------|--------------|
| A1  | shift(1) + shift(2) | Yes          |
| A2  | shift(1) + shift(2) | Yes          |
| A3  | shift(1) + shift(2) | Yes          |

All three report matching values and missing positions.  
No A run reportedly included assertions.  
Shared method does not establish identical full responses.

SOURCES: results-20260921.csv; trial-log-20260921.md; evidence-review-20260922.md

WHY CHOICES REMAIN OPEN

- 1

**A short request can succeed**  
All A reports describe shift(1) and shift(2). The source calls both SAS functions on every row.
- 2

**Check meaning as well as numbers**  
The first prior value is unavailable. Its missing value must remain missing after subtraction.
- 3

**Keep the sequence intact**  
Sorting, dropping rows or wrapping the final value into the first row changes the calculation.

ANTI-PATTERN: "FILL THE BLANKS"

Replacing the first missing prior value with zero manufactures a change of 10. Missing means there was no earlier observation.

Preserve missing positions and the five-row order.

# Prompt B: make the choices explicit

Low degrees of freedom / State the behavior, then ask for a check.

COPY THIS PROMPT + THE SAME SAS EXAMPLE

Convert this SAS code to Python using pandas.  
Show code in this response. Do not create or run files, or read existing Python files.

- Recreate id and value, preserving all five rows and their order.
- Each SAS LAG call executes on every row in this program.
- prev\_value: use value.shift(1), with no frequency argument.
- prev2\_value: use value.shift(2), with no frequency argument.
- change: subtract prev\_value from value.
- Leave unavailable prior values and resulting changes missing. Do not fill them with zero, drop rows, sort, or wrap values.
- Include assertions for the original id sequence and these lists:  
prev\_value: [missing, 10, 15, 12, 18]  
prev2\_value: [missing, missing, 10, 15, 12]  
change: [missing, 5, -3, 6, 2]  
Use pd.isna to test missing positions.
- Return one complete script. Compute results from value; do not hard-code the result columns.

Briefly explain why the first change is missing.

WHY THESE DETAILS HELP

- 1

**Protect the original sequence**  
Use row shifts without a frequency argument. Preserve all five ids and their order.
- 2

**Limit the equivalence**  
Shift fits these per-row calls. Conditional SAS LAG calls advance their queue only when executed.
- 3

**Make missing values checkable**  
B1-B3 report assertions and an explanation of the first missing change. Use pd.isna for missing positions.

REPORTED RESULTS / THREE B RUNS

| Run | Output match | Assertions |
|-----|--------------|------------|
| B1  | Yes          | Passed     |
| B2  | Yes          | Passed     |
| B3  | Yes          | Passed     |

CHECK IT YOURSELF  
**Row 3 change = -3; the first change stays missing.**  
Compare every value, missing position and original id.